

Seven Seas A-27

Security Audit

Jan 17, 2025

Version 1.0.0

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Introduction

This document includes the results of the security audit for Seven Seas's smart contract code as found in the section titled 'Source Code'. The security audit performed by the Macro security team lasted three days, from Jan 13th to Jan 17th, 2025.

The purpose of this audit is to review the source code of certain Seven Seas Solidity contracts, and provide feedback on the design, architecture, and quality of the source code with an emphasis on validating the correctness and security of the software in its entirety.

Disclaimer: While Macro's review is comprehensive and has surfaced some changes that should be made to the source code, this audit should not solely be relied upon for security, as no single audit is guaranteed to catch all possible bugs.

Overall Assessment

The following is an aggregation of issues found by the Macro Audit team:

Severity	Count	Acknowledged	Won't Do	Addressed
Medium	1	-	-	1

Seven Seas was quick to respond to these issues.

Specification

Our understanding of the specification was based on the following sources:

- Discussions with the Seven Seas team.
- Available documentation in the repository.

Trust Assumptions:

Some of the reviewed decoders allow Boring Vaults to interact with protocols controlled by multisig wallets that can update important state variables or upgrade the contract to a new version. It is trusted that the owners of this multisig will act in the interest of users. It should be understood that a loss of Boring Vault funds is possible if these wallets act maliciously.

Additionally, since the Silo V2 protocol is undergoing audits, boring vaults should not interact with the protocol until all audits are completed and issues are resolved.

Source Code

The following source code was reviewed during the audit:

- **Symbiotic Decoder (from [PR 48](#))**

Commit hash: `edfcba3f29f691c638b06d04f60af79503632b58`

Source Code	SHA256
<code>src/base/DecodersAndSanitizers/Protocols/SymbioticVaultDecoderAndSanitizer.sol</code>	<code>55949f2f9f3d1f0ea6d08fbdb19eb4dc5475669145c45dbcfac366a823d342d8</code>

- **BoringOnChainQueue capacity limit (from [PR 27](#))**

Commit Hash: `9b0ceb767fda7fd4d3001526189d7234e6f78142`

Source Code	SHA256
<code>src/base/Roles/BoringQueue/BoringOnChainQueue.sol</code>	<code>e446d88685f923ef28e301df6da66afa6fe19d7cc9dff104d798c362025ae5ba9</code>
<code>src/base/Roles/BoringQueue/BoringOnChainQueueWithTracking.sol</code>	<code>2de48f14cfd987c79366603905782f8683e46cb8f735a87790e0deb376b8ad6a</code>

- **Morpho Decoders (from [PR 54](#))**

Commit Hash: `79a5ced0fcc0eab61f7f8825c1cd9bef0668903d`

Source Code	SHA256
<code>src/base/DecodersAndSanitizers/MorphoRewardsDecoderAndSanitizer.sol</code>	<code>db9f5549ad3d1250f3aab04917de99ab5eccf76c7d799b433d53857a875e1d64</code>

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/MorphoRewardsMerkleClaimerDecoderAndSanitizer.sol	79621e1e7b77c64cd835c27167ab85bb4a15bb39e71912901c84302474130df4
src/base/DecodersAndSanitizers/Protocols/MorphoRewardsWrapperDecoderAndSanitizer.sol	ae91873a26455d31259648b3b6e32d90bc66279ad5d94f4f91d7638e119e0fbb

- **Teller Decoder (from [PR 61](#))**

Commit Hash: d580fd8bc9dd33c6857a40bd622464d4962de0ef

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/TellerDecoderAndSanitizer.sol	aace47c08c1b702758e89a1dc08e4256054b3afe4c1cbe82f54fdc05d78ac615
src/base/DecodersAndSanitizers/Protocols/WithdrawQueueDecoderAndSanitizer.sol	583a46847db78aef2d222c8c44420a92bf785858ae63198dbb26860ff55a9e04

- **Silo Decoder (from [PR 66](#))**

Commit Hash: 557277a8e6f91ddfa5804cc3b62281c6e253d470

Source Code	SHA256
src/base/DecodersAndSanitizers/Protocols/SiloDecoderAndSanitizer.sol	425b4ee7a8990af6e8f1d4cff1357e1889625b904e5a00e9c15f1d575f1980c6

Note: This document contains an audit solely of the Solidity contracts listed above. Specifically, the audit pertains only to the contracts themselves, and does not pertain

to any other programs or scripts, including deployment scripts.

Issue Descriptions and Recommendations

Click on an issue to jump to it, or scroll down to see them all.

M-1 Symbiotic decoder is unable to claim rewards

Security Level Reference

We quantify issues in three parts:

1. The high/medium/low/spec-breaking **impact** of the issue:

- How bad things can get (for a vulnerability)
- The significance of an improvement (for a code quality issue)
- The amount of gas saved (for a gas optimization)

2. The high/medium/low **likelihood** of the issue:

- How likely is the issue to occur (for a vulnerability)

3. The overall critical/high/medium/low **severity** of the issue.

This third part – the severity level – is a summary of how much consideration the client should give to fixing the issue. We assign severity according to the table of guidelines below:

Severity	Description
(C-x) Critical	We recommend the client must fix the issue, no matter what, because not fixing would mean significant funds/assets WILL be lost.
(H-x) High	We recommend the client must address the issue, no matter what, because not fixing would be very bad, or some funds/assets will be lost, or the code's behavior is against the provided spec.
(M-x) Medium	We recommend the client to seriously consider fixing the issue, as the implications of not fixing the issue are severe enough to impact the project significantly, albeit not in an existential manner.
(L-x) Low	The risk is small, unlikely, or may not be relevant to the project in a meaningful way. Whether or not the project wants to develop a fix is up to the goals and needs of the project.
(Q-x) Code Quality	The issue identified does not pose any obvious risk, but fixing could improve overall code quality, on-chain composability, developer ergonomics, or even certain aspects of protocol design.
(I-x) Informational	Warnings and things to keep in mind when operating the protocol. No immediate action required.
(G-x) Gas Optimizations	The presented optimization suggestion would save an amount of gas significant enough, in our opinion, to be worth the development cost of implementing it.

Issue Details

M-1

Symbiotic decoder is unable to claim rewards

TOPIC	STATUS	IMPACT	LIKELIHOOD
Protocol Design	Fixed ↗	Medium	Medium

SymbioticVaultDecoderAndSanitizer implements the `claimRewards` function from `IDefaultOperatorRewards` :

```
claimRewards(address recipient, address network, address token, uint256 totalC
```

Reference: [IDefaultOperatorRewards.sol#L83-L89](#)

However, since the boring vault functions as a protocol staker rather than an operator, it must implement the `claimRewards` function for stakers:

```
claimRewards(address recipient, address token, bytes calldata data)
```

Reference: [IStakerRewards.sol#L45](#)

Remediation to Consider

Update `claimRewards` function to be compatible with the implementation of `IStakerRewards.sol`.

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The scope of this report and review is limited to a review of only the code presented by the Seven Seas team and only the source code Macro notes as being within the scope of Macro's review within this report. This report does not include an audit of the deployment scripts used to deploy the Solidity contracts in the repository corresponding to this audit. Specifically, for the avoidance of doubt, this report does not constitute investment advice, is not intended to be relied upon as investment advice, is not an endorsement of this project or team, and it is not a guarantee as to the absolute security of the project. In this report you may through hypertext or other computer links, gain access to websites operated by persons other than Macro. Such hyperlinks are provided for your reference and convenience only, and are the exclusive responsibility of such websites' owners. You agree that Macro is not responsible for the content or operation of such websites, and that Macro shall have no liability to your or any other person or entity for the use of third party websites. Macro assumes no responsibility for the use of third party software and shall have no liability whatsoever to any person or entity for the accuracy or completeness of any outcome generated by such software.